Fundamental Trigonometric Identities

Section 1: Reciprocal and Quotient Identities

1 Simplify the expression $\tan x \cdot \csc x$.

a. $\sec x$

b. $\cos x$

c. $\sin x$

d. $\cot x$

2 Simplify $\frac{\cos x}{\cot x}$.

a. $\sin x$

b. $\cos^2 x$

c. $\tan x$

d. $\csc x$

3 If $\sin\theta = -3/5$ and $\tan\theta > 0$, find $\cos\theta$.

a. $-4/5$

b. $4/5$

c. $-3/4$

d. $5/4$

Section 2: Pythagorean Identities

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4 Which identity is directly derived by dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$?

a. $1 + \tan^2\theta = \sec^2\theta$

b. $1 + \cot^2\theta = \csc^2\theta$

c. $\sin^2\theta = 1 - \cos^2\theta$

d. $\tan^2\theta + \cot^2\theta = 1$

5 Simplify $(\sec\theta - 1)(\sec\theta + 1)$.

a. $\tan^2\theta$

b. $\cot^2\theta$

c. $\cos^2\theta$

d. 1

6 Simplify $\frac{1 - \cos^2 x}{\sin x}$.

a. $\sin x$

b. $\cos x$

c. $\csc x$

d. 1

Section 3: Even/Odd Identities and Symmetry

7 Which trigonometric functions are even functions ($f(-\theta) = f(\theta)$)?

a. $\cos\theta$ and $\sec\theta$

b. $\sin\theta$ and $\csc\theta$

c. $\tan\theta$ and $\cot\theta$

d. All six functions

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8 Simplify $\sin(-\theta) \cdot \cot(-\theta)$.

a. $\cos\theta$

b. $-\cos\theta$

c. $\sin\theta$

d. $-\sin\theta$

9 Simplify $\frac{\tan(-x)}{\sin(-x)}$.

a. $\sec x$

b. $-\sec x$

c. $\cos x$

d. $-\cos x$

Section 4: Verifying Identities Algebraically

10 Prove $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$. What technique verifies this algebraically?

a. Multiply numerator and denominator of the left side by the conjugate $(1-\cos x)$

b. Square both sides

c. Take the derivative

d. Cross multiply and set to 0

11 Simplify $\frac{\sec x - \cos x}{\sin x}$.

a. $\tan x$

b. $\cot x$

c. $\sec x$

d. $\sin x$

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12 Simplify $(1 + \tan^2\theta)(1 - \sin^2\theta)$.

a. 1

b. $\sec^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

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